

Capstone Reflection

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1. What did you get wrong?

The earliest scaffold treated functions/kk-payments as a Lambda, even though Week 9 already established kk-payments as the Kubernetes payments service. That naming collision would have failed a demo when reviewers asked which component writes to S3. The better approach - which we adopted - is a single producer in services/kk-payments/ and exactly three Lambdas in the receipt chain. I should have mapped Week 9 and Week 10 names on paper before creating directories.

2. Most important thing learned

Environment separation via Kustomize overlays (Week 9 ? capstone) changed how I think about "one manifest, many environments." Staging and production share k8s/base/deployment.yaml but differ only in ConfigMap patches (DB_HOST, RECEIPTS_BUCKET). That pattern appeared in Week 9 with a single namespace; the capstone forced the same idea across K8s namespaces and Serverless stages, which is the same design skill at two layers.

3. If you had a second pass, what would you add, remove, or change?

Change Component Why

Add IRSA on EKS instead of kk-payments-aws static keys Removes long-lived credentials from the cluster

Add DynamoDB idempotency table in kk-processor Prevent double-processing on S3 retry events

Remove Direct kubectl apply from Jenkins offline stage Use GitOps (Argo CD) so cluster state matches git

Change kk-analytics to incremental aggregates Listing all S3 keys on every event will not scale past demo volume

Submitted after live presentation. Honest assessment: AWS account activation was pending during build; offline validation and serverless package prove the pipeline, with cloud deploy as the final integration step.